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PHHP 36th Annual Research Day

Dynamic networks of methicillin-resistant *Staphylococcus aureus* in communities drive hospital transmission as revealed by whole-genome sequencing

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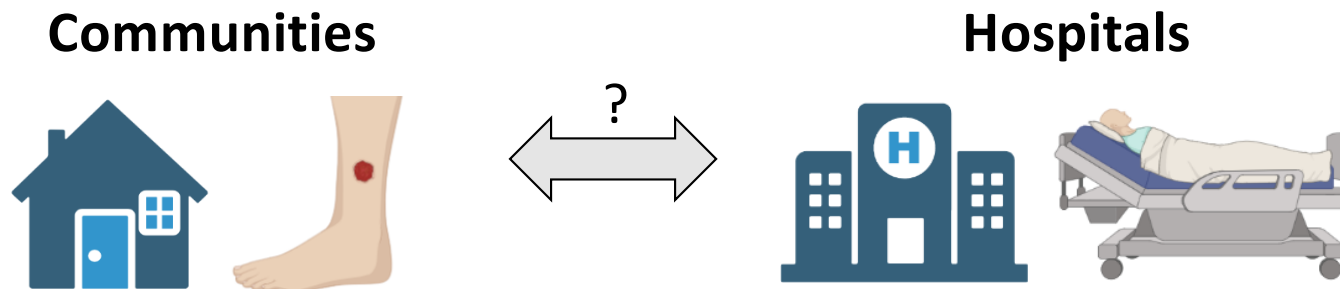
Department of Epidemiology

Mentors: Dr. Mattia Prosperi & Dr. J Glenn Morris

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BACKGROUND & SIGNIFICANCE

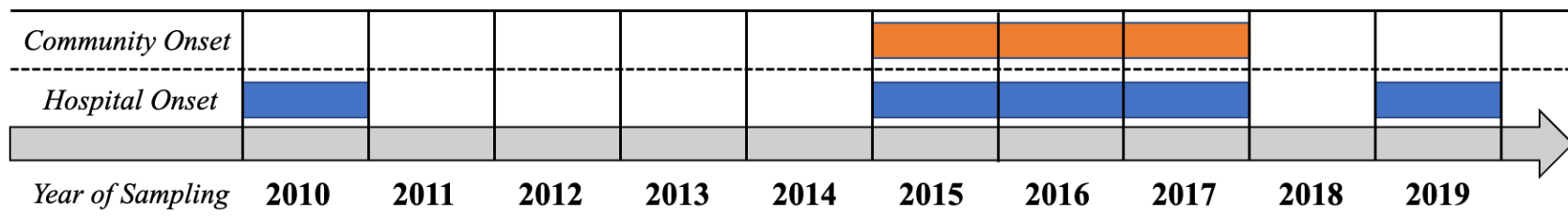
- Antibiotic resistance remains a global health threat with Methicillin-resistant Staphylococcus aureus (MRSA) leading the charge
- MRSA infections present in two main settings:



- Modern epidemiological analysis can elucidate the transmission pathways between the two settings and provide insight to prevention strategies

METHODS: Longitudinal Community Assessment

- **Community Sampling:** Patients presenting to UF Emergency Department with acute skin infection
- **Hospital Sampling:** Isolates collected from healthcare-acquired MRSA infections at UF Health



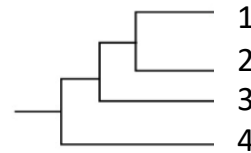
Epidemiological Analysis

Case Investigation
Spatial Visualization

+

Whole Genome Sequencing

1. ACCGUA
2. ACCGAA
3. ACCGCA
4. ACCAUA

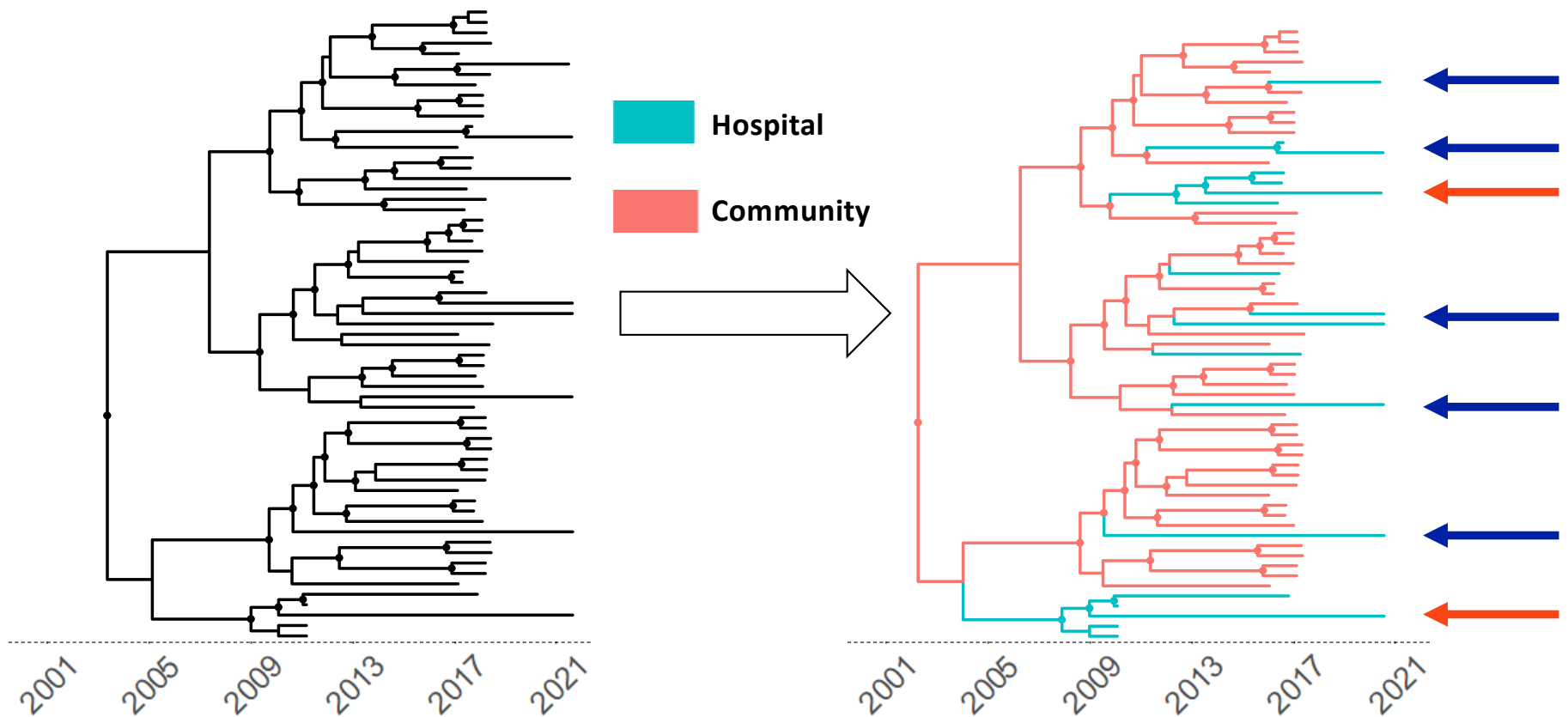


Questions:

Transmission Patterns?
Source Tracing?
Risk Factors?

RESULTS: Community-Hospital Transmission

Time of Infection + Genome + Predictable Mutation Rate = **Time-Scaled Phylogenetic Tree**



CONCLUSION

1. Intra-hospital MRSA outbreaks can be sustained for at least 10 years
2. Hospital MRSA origins can largely be tracked to community strains
3. Little evidence to suggest the reverse

Nosocomial MRSA outbreak prevention strategies should target unique aspects of the community rather than focusing solely on the hospital



Authors & Contributors (+ More!)

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